

PFM -- Personal Finance Manager

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Presentation Outline

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Introduction

- What is Personal Finance Manager?
- Why expense tracking is important
- Problem with traditional manual finance apps

Problem Statement

- Manual data entry is repetitive
- Users abandon tracking over time
- Existing apps lack smart categorization
- Need for intelligent budget analysis

Project Objectives

To develop a smart Personal Finance Manager that automates expense tracking and budget analysis using natural language processing.

Scope and Limitations

Scope

- Expense tracking
- NLP-based categorization
- Budget optimization
- Dashboard analytics
- RAG-based financial assistant

Limitations

- No bank integration
- No receipt scanning
- No tax calculation
- No multi-currency support

Background Study

- Existing finance apps depend on manual input
- PFM uses NLP, RAG, and optimization
- Full-stack web application using React, FastAPI, and Supabase

Literature Review

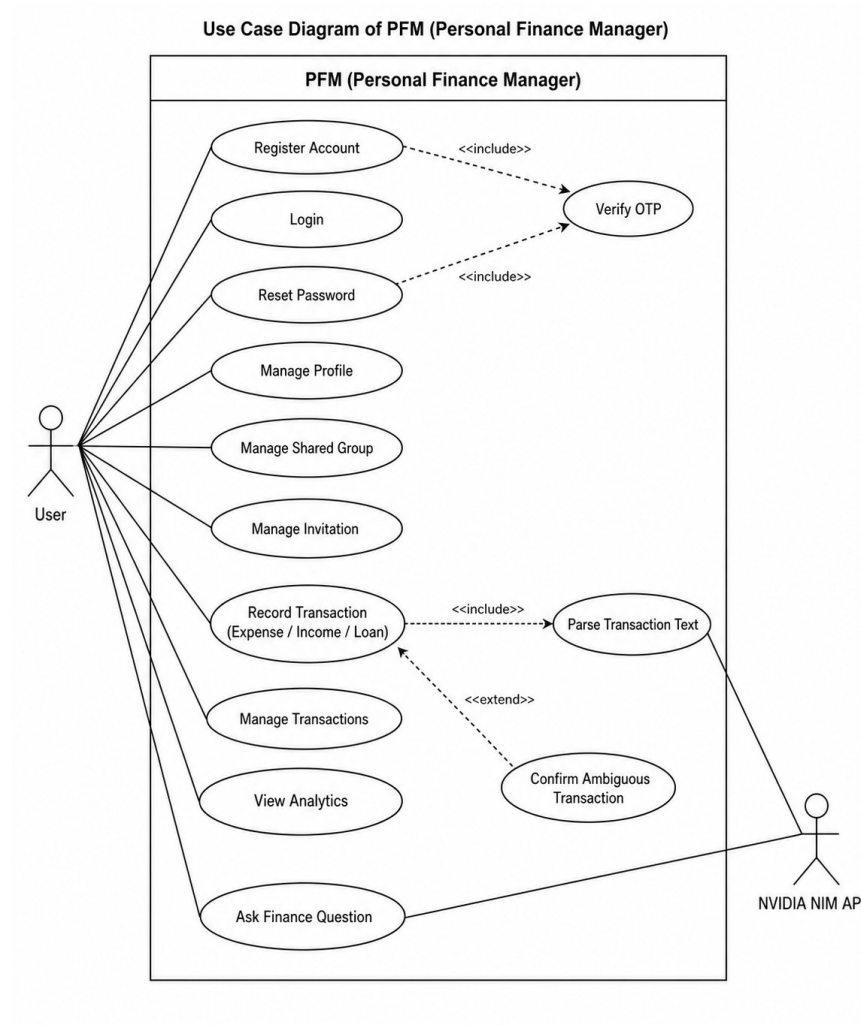
- Splitwise
- YNAB
- Wallet by BudgetBakers
- Automated text categorization
- Retrieval-Augmented Generation

Agile Methodology

- Planning
- Design
- Development
- Testing
- Deployment
- Review and Maintenance

FIGURE 3.2: USE CASE DIAGRAM OF PFM

Use Case Diagram



Mention

- User registration
- Login
- Manage profile
- Record transaction
- Manage transactions
- View analytics
- Finance question
- NVIDIA NIM API interaction

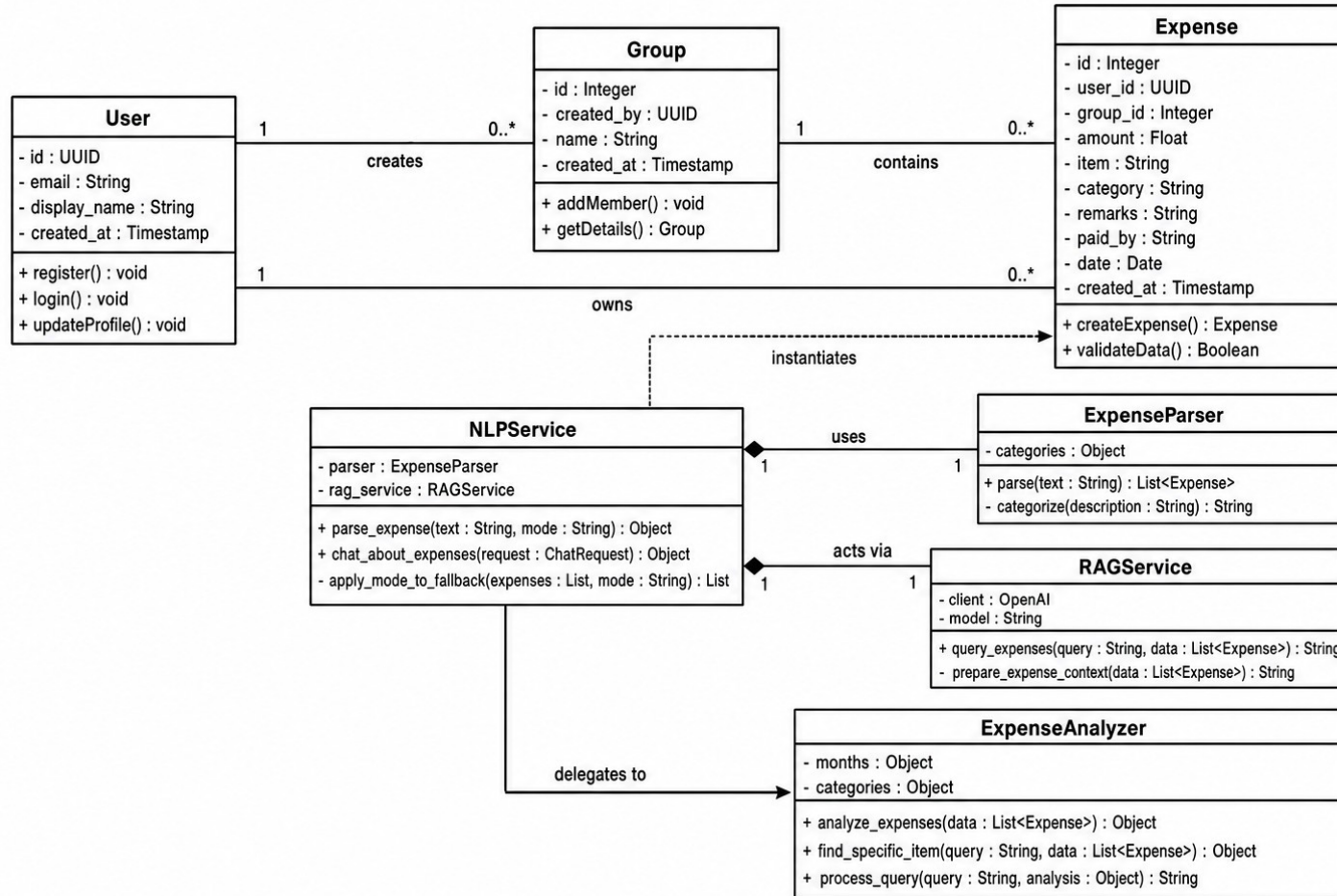
Project Timeline

- Project planning timeline
- Designing phase
- Development phase
- Testing phase
- Deployment phase
- Review and maintenance phase

FIGURE 3.4: CLASS DIAGRAM OF PFM

Class Diagram

Class Diagram of PFM (Personal Finance Manager)



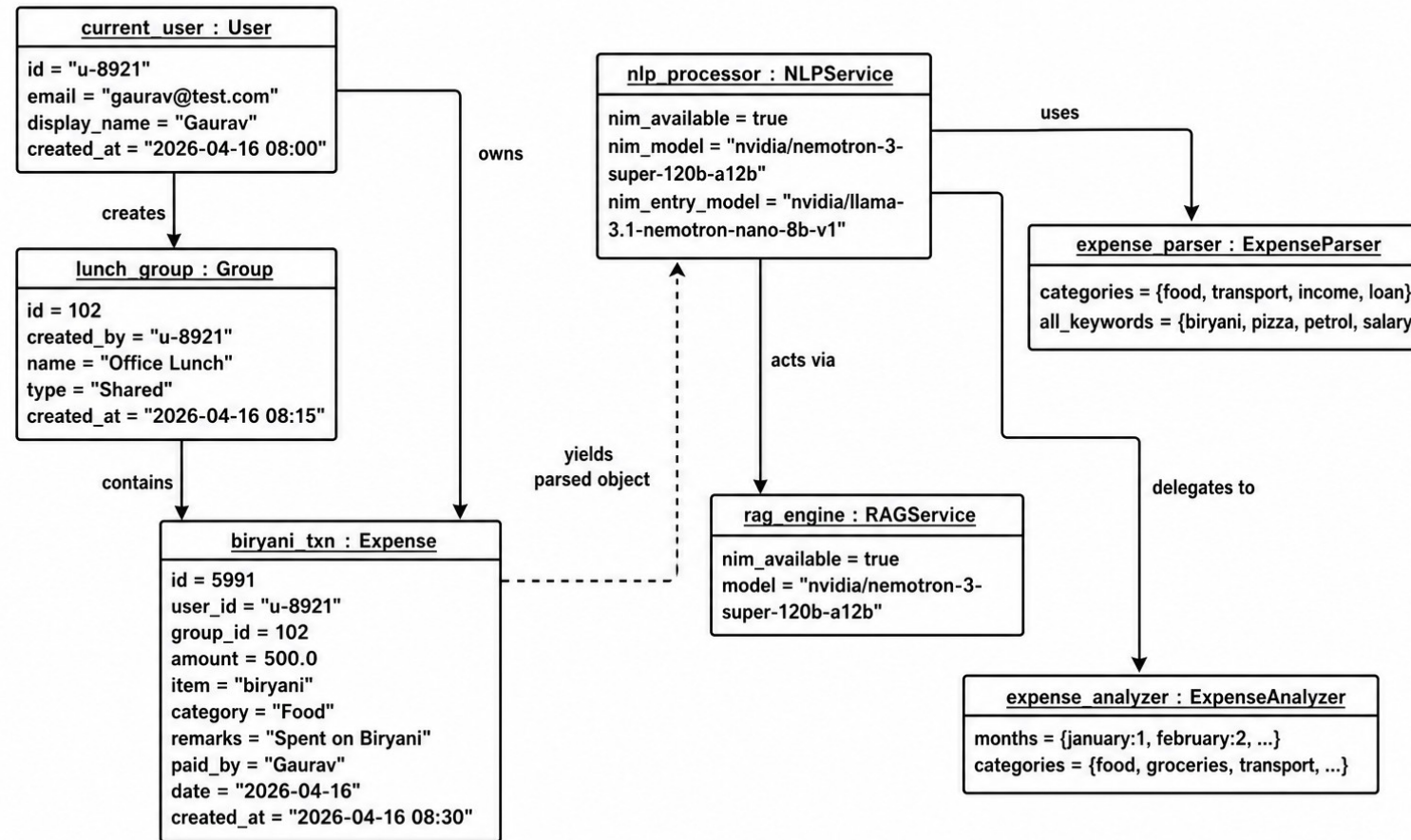
Mention

- User class
- Group class
- Expense class
- NLP Service
- Expense Parser
- RAG Service
- Expense Analyzer
- Relationships between system components

FIGURE 3.5: OBJECT DIAGRAM OF PFM

Object Diagram

Object Diagram of PFM (Personal Finance Manager)



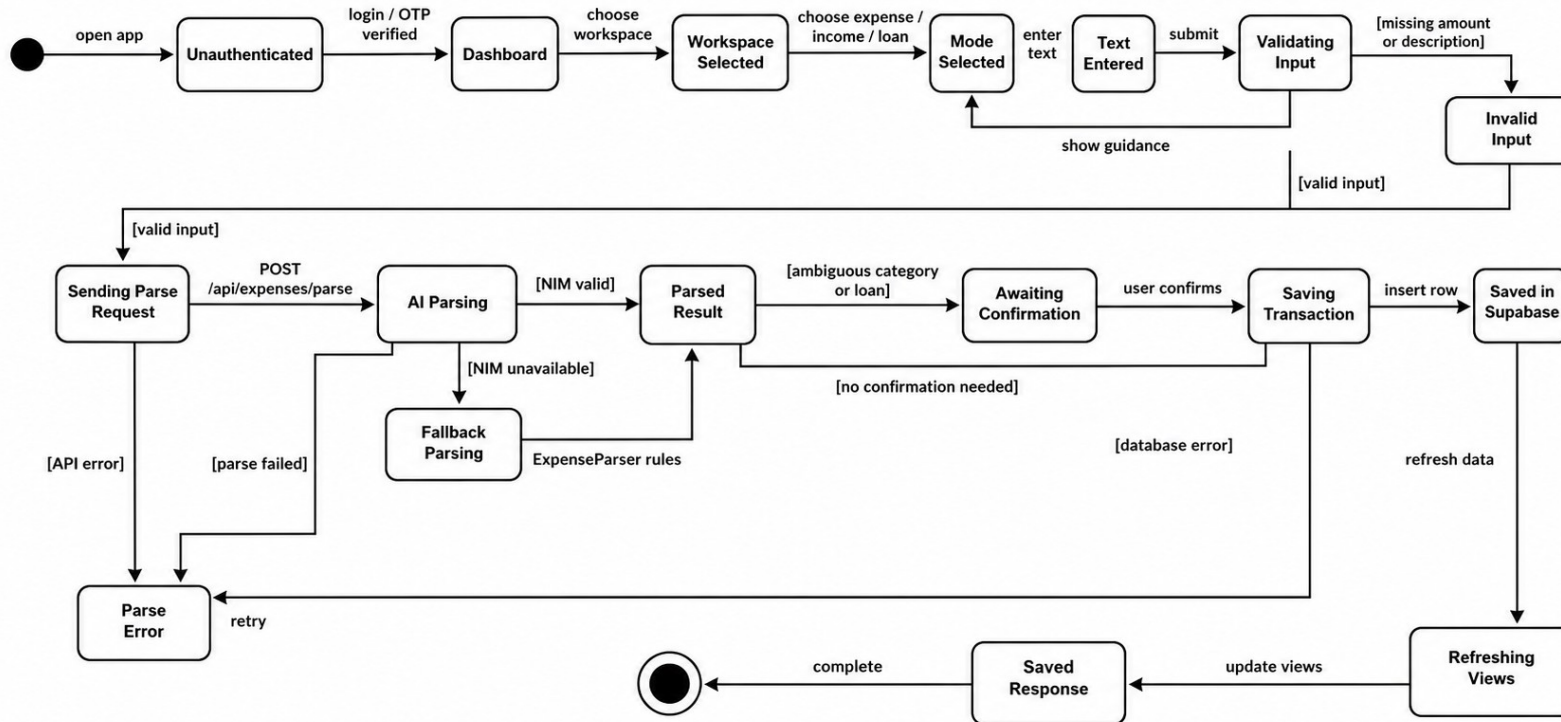
Mention

- Runtime example of system objects
- Current user object
- Group object
- Expense object
- NLP service object
- RAG engine object
- Expense parser and analyzer objects

FIGURE 3.6: STATE DIAGRAM OF PFM

State Diagram

State Diagram of PFM Transaction Recording

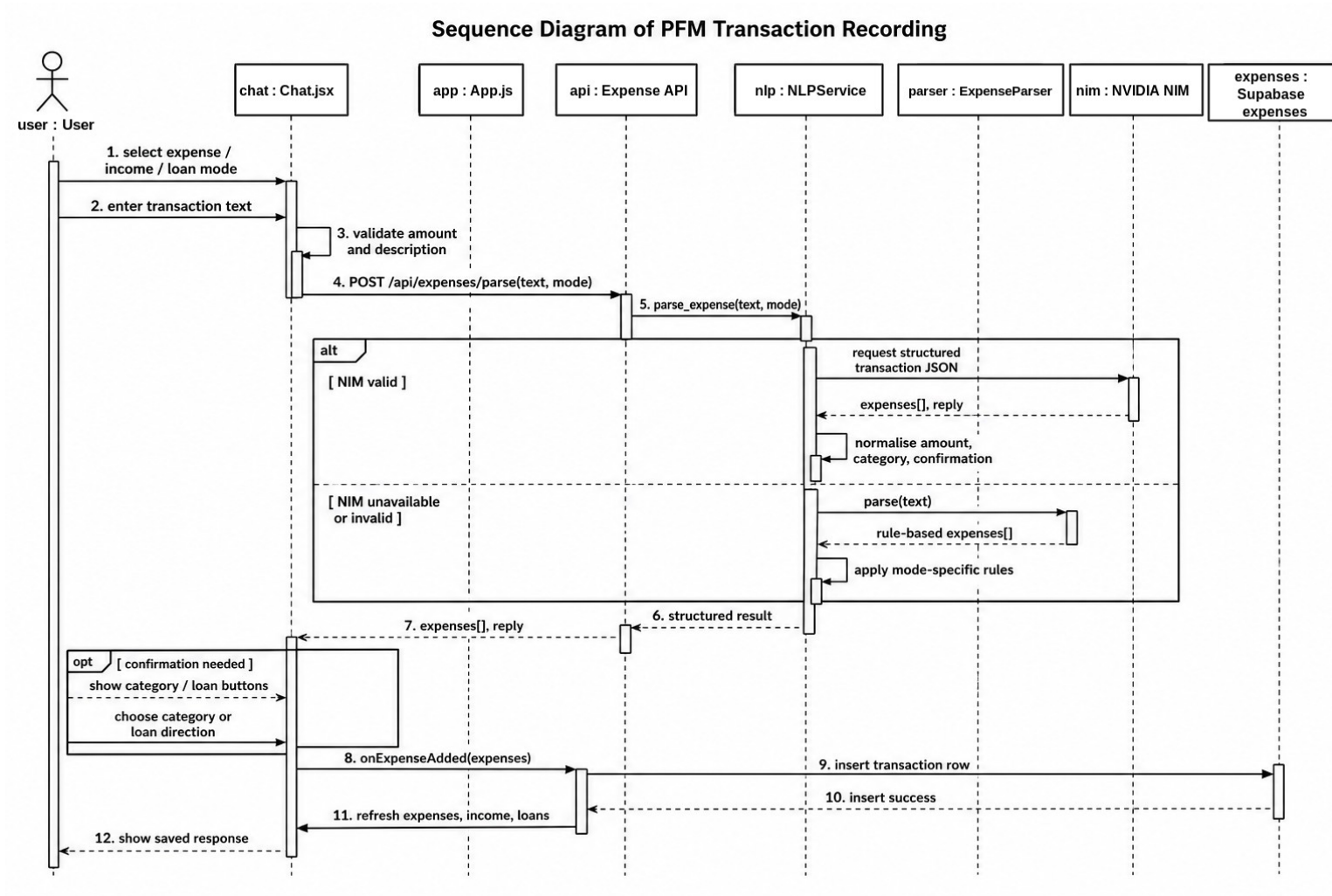


Mention

- User authentication state
- Dashboard state
- Transaction input state
- Parsing state
- Confirmation state
- Saving transaction state
- Response and refresh state

FIGURE 3.7: SEQUENCE DIAGRAM OF PFM

Sequence Diagram

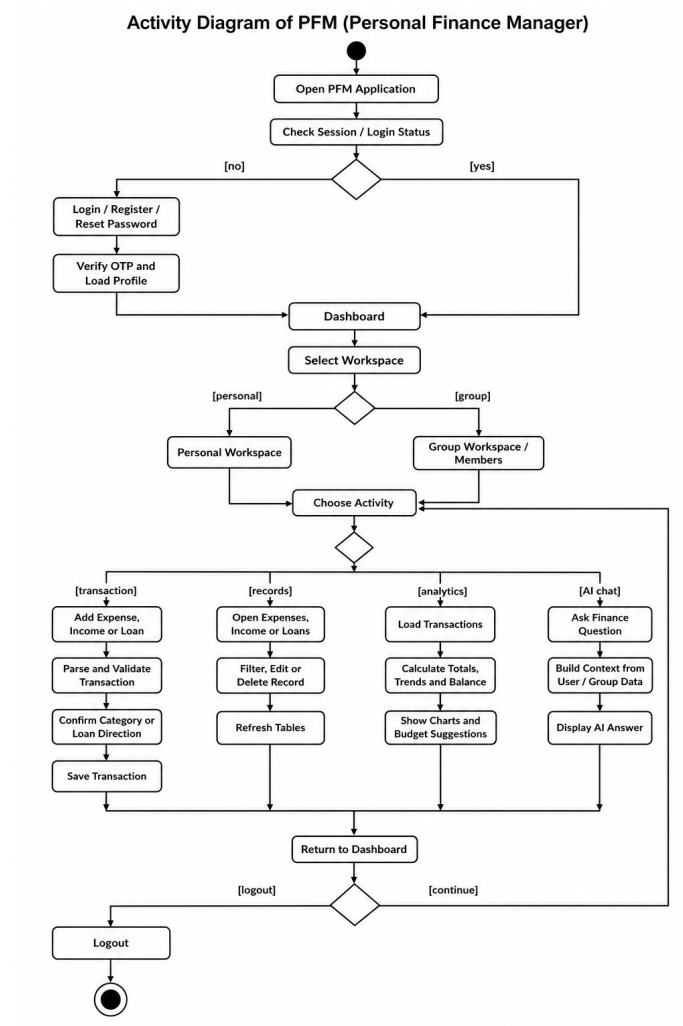


Mention

- User enters transaction text
- Chat.js sends request
- App.js validates input
- Expense API communicates with NLP Service
- Expense Parser or NVIDIA NIM processes input
- Supabase stores transaction
- Response is shown to user

FIGURE 3.8: ACTIVITY DIAGRAM OF PFM

Activity Diagram

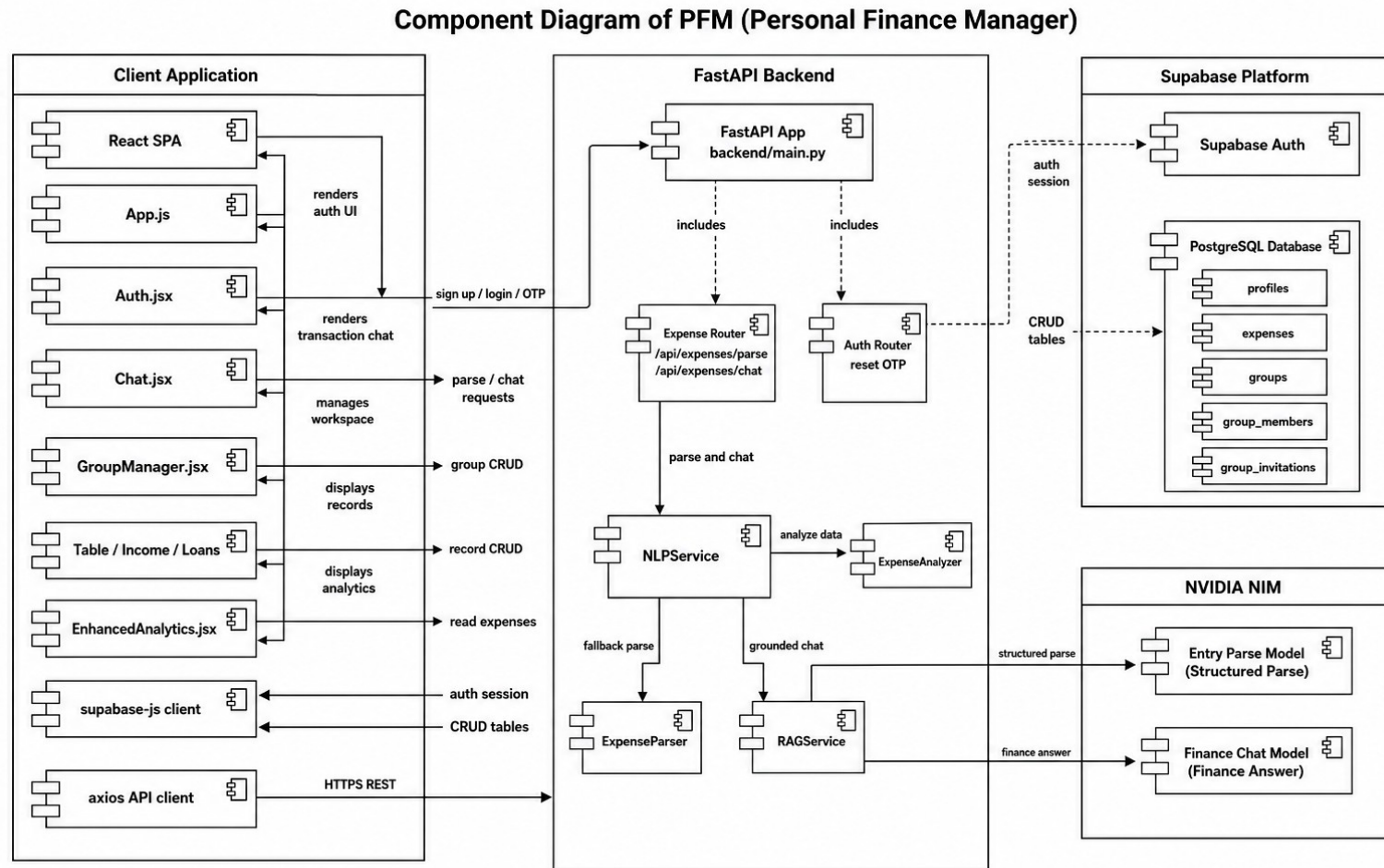


Mention

- Application start
- Authentication check
- Dashboard display
- Record transaction flow
- Review/AI flow
- Ask AI question flow
- Save transaction
- Logout

FIGURE 3.9: COMPONENT DIAGRAM OF PFM

Component Diagram



Mention

- Client application
- FastAPI backend
- Supabase platform
- PostgreSQL database
- NVIDIA NIM API
- NLP Service
- RAG Service
- Expense Analyzer

Algorithm Details

- Convert input text to lowercase
- Extract amount using regex
- Match keywords with expense categories
- Select category with highest score
- Return amount and category

Implementation Tools

- Frontend: React.js, Tailwind CSS, Recharts
- Backend: FastAPI
- Database: Supabase PostgreSQL
- AI/API: NIM API
- CASE Tools: VS Code, Draw.io, Lucidchart, Figma

Implementation Details of Modules

- Registration Module
- Login Module
- Expense Management Module
- NLP Expense Parsing Module
- Budget Optimization Module
- Dashboard and Analytics Module
- Conversational Financial Assistant Module

Testing

- User registration
- User login
- Add expense
- NLP expense parsing
- Budget optimization
- Dashboard analytics
- Conversational query
- Delete expense
- Logout

Modules Completed

- User Authentication
- Expense Management Module
- NLP Expense Parsing Module
- Dashboard and Analytics Module
- Budget Optimization Module
- Frontend Development
- Backend API Development
- Database Integration

Modules Remaining

- Conversational Financial Assistant / RAG
- Gemini API Optimization
- System Testing and Debugging
- Deployment and Performance Optimization
- Report Documentation and Final Presentation

Conclusion

- PFM reduces manual expense entry
- NLP makes tracking faster and easier
- Dashboard helps users understand spending
- Budget optimization supports better saving decisions
- Remaining work focuses on RAG, testing, deployment, and final documentation

References

- Splitwise
- YNAB
- Wallet by BudgetBakers
- Automated Text Categorization
- Retrieval-Augmented Generation